

# Medvale Internal Medicine

## Chapter 2B-2C: Obstructive Lung Disease

COPD, Asthma, Bronchiectasis, and Adult Cystic Fibrosis Patterns

Purpose. This chapter is written as an original internal medicine study resource. The uploaded sample pages and Anki card export were used only as coverage signals to choose topics and ensure high-yield breadth; no source media or protected figure layouts are reproduced. All diagrams and tables in this chapter are newly created for Medvale.

RAG design note. The prose intentionally repeats concept relationships in paragraph form - mechanism, presentation, diagnosis, management, and complications - so a graph RAG can extract nodes such as COPD, emphysema, chronic bronchitis, asthma, bronchiectasis, cystic fibrosis, FEV1/FVC, hyperinflation, hypoxemia, oxygen therapy, and exacerbation treatment.

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# 1. Why obstructive lung disease matters clinically

Obstructive lung diseases are disorders in which air leaves the lungs too slowly. The central physiologic abnormality is airflow limitation, usually seen on spirometry as a reduced FEV1/FVC ratio. In practical internal medicine, this single physiologic pattern appears in several different clinical scripts: a 68-year-old smoker with chronic dyspnea and hyperinflated lungs; a 24-year-old with episodic wheezing after viral illness or exercise; a 51-year-old with years of purulent sputum and recurrent infections; and an adult with cystic fibrosis who has chronic *Pseudomonas* airway infection, pancreatic insufficiency, and bronchiectasis. The exam task is to recognize which script is active, determine acuity, and identify when respiratory failure is developing.

The core diseases in this segment are chronic obstructive pulmonary disease (COPD), asthma, bronchiectasis, and adult cystic fibrosis. They overlap because mucus plugging, bronchospasm, airway wall inflammation, and air trapping can all produce wheezing and dyspnea. They differ in their dominant driver: COPD is usually exposure-related fixed airflow limitation; asthma is variable airway inflammation and hyperresponsiveness; bronchiectasis is chronically dilated, damaged bronchi with impaired secretion clearance; cystic fibrosis is a multisystem chloride channel disorder that commonly produces bronchiectasis and recurrent infection.

**High-yield numbers to anchor the chapter: post-bronchodilator FEV1/FVC < 0.70 supports persistent airflow obstruction; an asthma bronchodilator response is often defined as an increase in FEV1 or FVC of at least 12% and at least 200 mL; long-term oxygen therapy in COPD is classically indicated for PaO2 ≤ 55 mm Hg or SaO2 ≤ 88%, or PaO2 56-59 mm Hg with cor pulmonale, pulmonary hypertension, or erythrocytosis; during acute COPD exacerbation, oxygen is usually titrated to SpO2 88-92% to avoid both hypoxemia and hyperoxia-associated hypercapnia.**

## 2. Core physiology: obstruction, air trapping, and gas exchange

In obstructive disease, the earliest bedside clue is often an expiratory problem. Patients may say they cannot “get air out,” and the clinician may hear prolonged expiration, wheeze, or quiet breath sounds. Quiet breath sounds can be more ominous than loud wheeze, because it may mean very low airflow. The spirometric pattern is a reduced FEV1/FVC ratio because FEV1 falls disproportionately. Total lung capacity and residual volume may rise when air trapping and hyperinflation are present. Diffusing capacity helps separate emphysema, in which alveolar surface area is destroyed and DLCO often falls, from asthma or chronic bronchitis, where DLCO is often normal unless another process coexists.

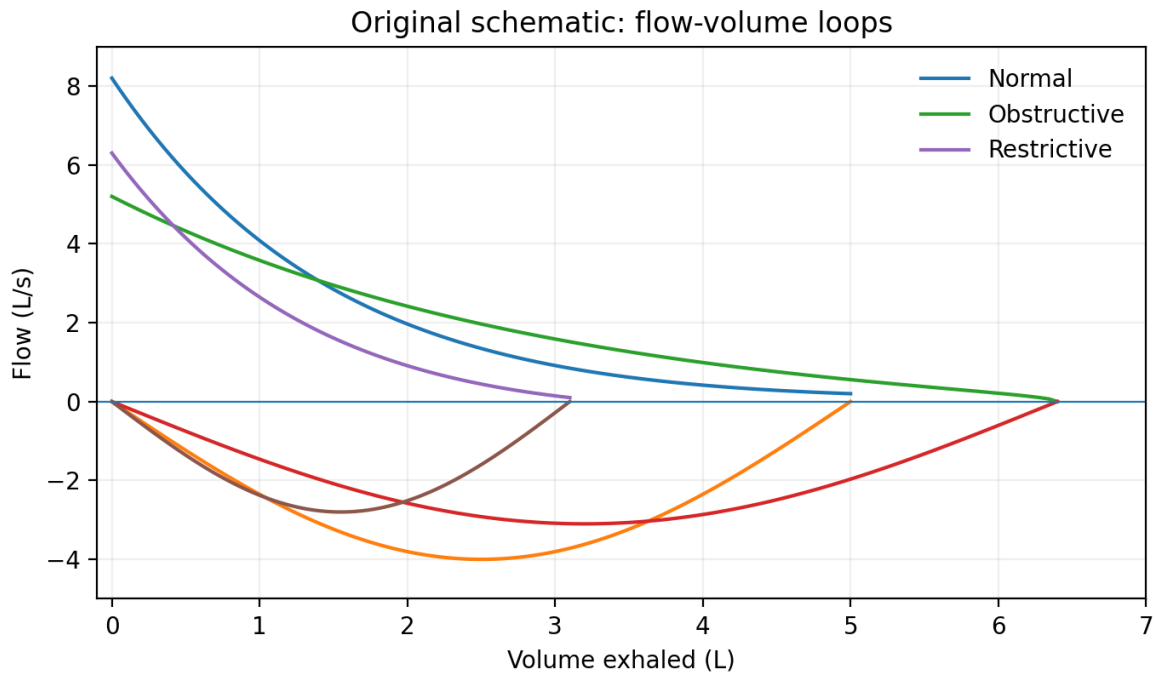


Figure 2B-1. Original schematic of flow-volume loops. Obstruction produces a scooped expiratory limb and often larger absolute lung volumes; restriction produces smaller volumes with preserved or increased FEV1/FVC.

| Measurement     | Obstructive pattern                                         | Restrictive pattern                              | Clinical interpretation                                                                                                                  |
|-----------------|-------------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| FEV1            | Low                                                         | Low or normal                                    | Low FEV1 reflects reduced forced expiratory flow; severity grading in COPD uses percent predicted FEV1 after obstruction is established. |
| FVC             | Normal or low                                               | Low                                              | FVC may fall in severe obstruction because air trapping prevents full exhalation.                                                        |
| FEV1/FVC        | Low; often <0.70                                            | Normal or high                                   | Most useful first discriminator between obstruction and restriction.                                                                     |
| TLC             | Normal or high                                              | Low                                              | High TLC suggests hyperinflation; low TLC confirms restriction.                                                                          |
| Residual volume | High when air trapping is present                           | Low, normal, or high depending on cause          | High RV explains barrel chest, flattened diaphragms, and dynamic hyperinflation.                                                         |
| DLCO            | Low in emphysema; often normal in asthma/chronic bronchitis | Low in ILD; normal in extrapulmonary restriction | DLCO connects physiology to alveolar-capillary surface area.                                                                             |

### 3. COPD: illness script and pathophysiology

COPD is a chronic respiratory syndrome defined by persistent respiratory symptoms and airflow limitation, most often after years of exposure to tobacco smoke, biomass fuel, or occupational dusts and fumes. It is not just “smoker with shortness of breath.” COPD is a progressive inflammatory injury pattern that narrows small airways, destroys elastic recoil, increases mucus production, and creates air trapping. The patient experiences exertional dyspnea because each breath begins before the previous breath has fully emptied, especially during walking, stairs, infection, or tachypnea.

Two classic pathologic phenotypes help organize thinking. Chronic bronchitis is a clinical syndrome of productive cough for at least 3 months per year for at least 2 consecutive years. It reflects mucus gland enlargement, goblet cell hyperplasia, airway inflammation, and recurrent infection or colonization. Emphysema is an anatomic syndrome of permanent enlargement of airspaces distal to terminal bronchioles with destruction of alveolar walls. Centrilobular emphysema is strongly associated with cigarette smoking and tends to involve upper lobes; panacinar emphysema is associated with alpha-1 antitrypsin deficiency and tends to involve lower lobes.

COPD mechanism map: exposure to clinical decompensation

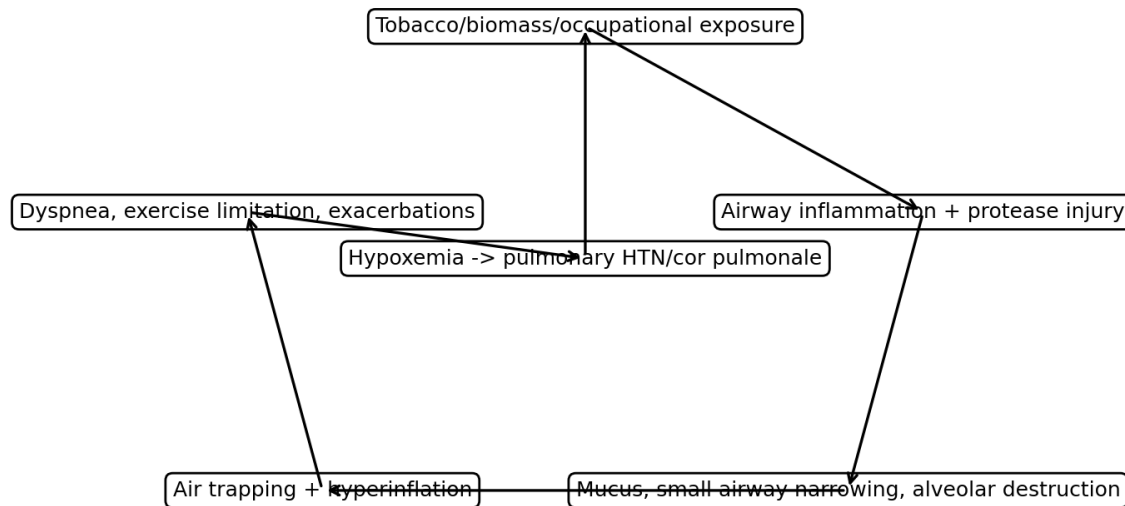


Figure 2B-2. COPD mechanism map. The key RAG relationship is exposure -> inflammation/protease injury -> airway narrowing and alveolar destruction -> air trapping -> dyspnea and exacerbations -> hypoxemic complications.

Alpha-1 antitrypsin normally restrains neutrophil elastase. When alpha-1 antitrypsin is deficient, elastase activity destroys alveolar walls more easily, especially when smoking simultaneously recruits neutrophils and oxidatively inactivates remaining antiprotease activity. Consider testing in COPD that begins before age 45, emphysema in a nonsmoker, basilar-predominant emphysema, family history, or unexplained liver disease.

## 4. COPD presentation patterns

The typical COPD presentation is slowly progressive dyspnea, reduced exercise tolerance, chronic cough, and sputum production. Symptoms may be subtle because patients unconsciously reduce activity over years. Physical examination may show tachypnea, prolonged expiratory phase, pursed-lip breathing, accessory muscle use, decreased breath sounds, wheezing, barrel chest, hyperresonance, and distant heart sounds. In advanced disease, chronic hypoxemia can cause pulmonary hypertension, cor pulmonale, peripheral edema, secondary polycythemia, and exertional syncope.

| Feature          | Chronic bronchitis-predominant COPD                                   | Emphysema-predominant COPD                                              |
|------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------|
| Dominant symptom | Daily productive cough, sputum, recurrent infectious exacerbations    | Progressive exertional dyspnea, low exercise reserve                    |
| Pathology        | Mucus gland enlargement, goblet cell hyperplasia, airway inflammation | Alveolar wall destruction, reduced elastic recoil, airspace enlargement |

| Feature            | Chronic bronchitis-predominant COPD                    | Emphysema-predominant COPD                                                                         |
|--------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Gas exchange       | Hypoxemia and hypercapnia may appear earlier           | DLCO often reduced; hypercapnia may appear late                                                    |
| Exam clue          | Rhonchi, wheeze, edema/cor pulmonale in severe disease | Thin habitus, pursed-lip breathing, hyperinflated chest                                            |
| Chest imaging clue | Prominent bronchovascular markings may be seen         | Hyperinflation, flattened diaphragms, increased retrosternal airspace, decreased vascular markings |

**Common mimics include heart failure, anemia, deconditioning, interstitial lung disease, chronic thromboembolic disease, vocal cord dysfunction, obesity hypoventilation, and medication effects such as beta-blocker-induced bronchospasm. A COPD label should not stop the clinician from considering pulmonary embolism, pneumonia, pneumothorax, acute coronary syndrome, or arrhythmia when symptoms change abruptly.**

## 5. COPD diagnostic approach

Spirometry is required to confirm COPD. The teaching threshold is post-bronchodilator FEV1/FVC < 0.70, which indicates persistent airflow obstruction. COPD severity is then partly described by FEV1 percent predicted, but treatment decisions should also use symptom burden, exacerbation history, blood eosinophil count when considering inhaled corticosteroids, smoking status, oxygenation, and comorbid disease. A chest x-ray is not diagnostic for mild COPD but can show hyperinflation in advanced emphysema and is useful during acute changes to look for pneumonia, pulmonary edema, pneumothorax, or malignancy. CT can characterize emphysema distribution and evaluate lung cancer screening findings or alternative diagnoses.

| Test                                   | What it answers                                | Numbers or interpretation                                                                         |
|----------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Spirometry before/after bronchodilator | Is there obstruction? Is it reversible?        | COPD: post-bronchodilator FEV1/FVC <0.70. Asthma support: FEV1 or FVC increases ≥12% and ≥200 mL. |
| Lung volumes                           | Is there hyperinflation or restriction?        | High RV/TLC suggests air trapping. Low TLC confirms restriction.                                  |
| DLCO                                   | Is alveolar-capillary surface area impaired?   | Low in emphysema, pulmonary vascular disease, ILD, anemia. Often normal in asthma.                |
| Pulse oximetry/ABG                     | Is there hypoxemia or hypercapnia?             | ABG is useful in severe exacerbation, altered mental status, or concern for CO2 retention.        |
| CBC                                    | Is there infection, anemia, or erythrocytosis? | Eosinophils may influence ICS benefit; hematocrit may rise with chronic hypoxemia.                |
| Chest imaging                          | Are there complications or mimics?             | CXR in exacerbation; CT for unclear diagnosis, emphysema distribution, bronchiectasis, nodules.   |

## 6. COPD stable management

The most important disease-modifying intervention is smoking cessation and exposure removal. Stopping smoking slows the accelerated decline in FEV1; it does not usually restore lung function to that of a never-smoker, but it changes the slope of future loss. Vaccination, physical activity, pulmonary rehabilitation, inhaler technique, nutrition, and comorbidity management are as important as inhalers because COPD outcomes are driven by exacerbations, cardiovascular disease, frailty, and functional reserve.

Bronchodilators are the foundation of symptom control. Short-acting beta agonists and short-acting muscarinic antagonists provide quick relief. Long-acting beta agonists and long-acting muscarinic antagonists are maintenance agents; dual LABA/LAMA therapy is often used when symptoms persist. Inhaled corticosteroids are not universal COPD drugs. They are considered when exacerbation risk is high, particularly with higher blood eosinophils or asthma-COPD overlap, but they increase pneumonia risk and should be individualized. Oral steroids are not chronic maintenance therapy for stable COPD.

| Intervention               | Primary role                                                 | Exam/RAG association                                                                                              |
|----------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Smoking cessation          | Slows FEV1 decline and reduces mortality risk                | Most important intervention for smokers with COPD.                                                                |
| Long-acting bronchodilator | Improves symptoms, exercise tolerance, and exacerbation risk | LAMA or LABA; combine LABA/LAMA if persistent symptoms.                                                           |
| Inhaled corticosteroid     | Reduces exacerbations in selected patients                   | Consider with frequent exacerbations/eosinophilic phenotype; pneumonia risk.                                      |
| Pulmonary rehabilitation   | Improves dyspnea, function, and quality of life              | Best nonpharmacologic option for functional limitation.                                                           |
| Long-term oxygen           | Improves survival in severe chronic resting hypoxemia        | PaO <sub>2</sub> ≤55 mm Hg or SaO <sub>2</sub> ≤88%; also PaO <sub>2</sub> 56-59 with cor pulmonale/polycythemia. |
| Vaccination                | Prevents respiratory infection triggers                      | Annual influenza; pneumococcal per age/risk guidance; COVID/RSV as appropriate.                                   |

## 7. COPD exacerbation: recognition and management

A COPD exacerbation is an acute worsening of dyspnea, cough, and/or sputum that requires additional therapy. Triggers include viral infection, bacterial infection, air pollution, medication nonadherence, pulmonary embolism, heart failure, pneumothorax, and sedative use. Increased sputum purulence suggests bacterial involvement, particularly with increased sputum volume and dyspnea. The first task is severity triage: respiratory rate, accessory muscle use, mental status, oxygen saturation, ability to speak, hemodynamics, and signs of fatigue.

Initial treatment includes short-acting bronchodilators, systemic corticosteroids, controlled oxygen, and antibiotics when indicated. Oxygen is titrated rather than maximized; a common target is SpO<sub>2</sub> 88-92% in COPD exacerbation because excessive oxygen can worsen hypercapnia by worsening V/Q mismatch, reducing hypoxic pulmonary vasoconstriction, and altering hemoglobin-CO<sub>2</sub> carriage. Noninvasive ventilation is strongly considered when acute or acute-on-chronic hypercapnic respiratory failure causes respiratory acidosis, severe dyspnea, or increased work of breathing. Intubation is needed for inability to protect the airway, severe encephalopathy, worsening acidosis despite NIV, shock, or exhaustion.

| Step              | Action                                                                                                | Why it matters                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| 1. Triage         | Assess RR, SpO <sub>2</sub> , mental status, work of breathing, BP, CXR if needed                     | Identifies pneumonia, pneumothorax, pulmonary edema, and impending respiratory failure.                  |
| 2. Bronchodilator | SABA plus SAMA initially; then reassess airflow and distress                                          | Albuterol opens smooth muscle; ipratropium adds anticholinergic bronchodilation.                         |
| 3. Steroids       | Short systemic course for moderate/severe exacerbation                                                | Reduces treatment failure and speeds recovery; avoid chronic systemic use.                               |
| 4. Antibiotics    | Use when sputum purulence plus increased dyspnea/volume, severe exacerbation, or ventilation required | Targets common bacteria such as H. influenzae, S. pneumoniae, M. catarrhalis; consider Pseudomonas risk. |

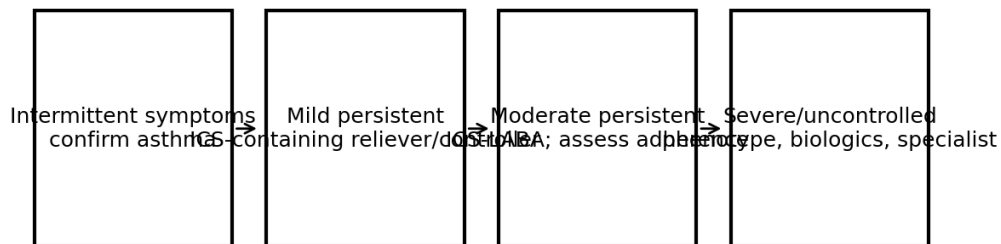
| Step                   | Action                                                                               | Why it matters                                                        |
|------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 5. Oxygen              | Controlled O2, often SpO2 88-92%                                                     | Prevents hypoxemia while limiting hyperoxia-induced hypercapnia.      |
| 6. Ventilatory support | NIV for hypercapnic acidosis; intubate if NIV fails or patient cannot protect airway | Reduces work of breathing and buys time for bronchodilators/steroids. |

## 8. Asthma: illness script and mechanism

Asthma is a chronic airway disease characterized by variable respiratory symptoms, variable expiratory airflow limitation, airway inflammation, and airway hyperresponsiveness. The word variable is essential: asthma symptoms fluctuate over time and intensity. Patients may have wheeze, cough, chest tightness, or shortness of breath that worsens at night, early morning, with viral infection, allergens, exercise, cold air, irritants, aspirin/NSAIDs in susceptible patients, or occupational exposures. A normal exam between attacks does not exclude asthma.

Asthma inflammation makes the airway twitchy. Allergen or irritant exposure activates mast cells, eosinophils, T helper 2 pathways, epithelial cytokines, and smooth muscle contraction. Narrow airways then create wheeze and air trapping. With time, uncontrolled asthma can cause airway remodeling, mucus plugging, and a less reversible component. However, compared with COPD, asthma is often more reversible and may begin at any age. Adult-onset asthma should prompt attention to occupational triggers, aspirin-exacerbated respiratory disease, eosinophilic disease, obesity, reflux, rhinitis, and medication exposures.

Asthma control framework: step up, reassess, step down when safe



At every step: inhaler technique, triggers, comorbid rhinitis/GERD/OSA, smoking/vaping, action plan.

Figure 2C-1. Asthma control framework. Step-up therapy should be paired with adherence checks, trigger control, inhaler technique, and reassessment.

## 9. Asthma diagnosis and severity assessment

Asthma diagnosis requires a compatible history plus evidence of variable airflow limitation when possible. Spirometry may show obstruction with reduced FEV1/FVC and reversibility after bronchodilator. A common adult threshold for significant bronchodilator response is an increase in FEV1 or FVC of at least 12% and at least 200 mL. If baseline spirometry is normal but suspicion remains high, peak flow variability, bronchoprovocation testing, or repeating spirometry during



symptoms can be useful. Methacholine challenge supports airway hyperresponsiveness when it produces a fall in FEV1 at low dose, but it is not used during severe acute symptoms.

Asthma severity can be described before treatment by symptom frequency, nighttime awakening, rescue inhaler use, activity limitation, and lung function. Control is assessed after treatment by current symptoms, exacerbations, medication side effects, inhaler technique, adherence, and risk factors. The patient who has required prior intubation or ICU care for asthma has a major risk factor for future life-threatening asthma, even if today's symptoms seem modest.

| Clue                                                | Meaning                                       | Pitfall                                                                                                   |
|-----------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Wheeze, cough, chest tightness vary over time       | Supports asthma script                        | Not all wheeze is asthma; CHF, COPD, PE, anaphylaxis, foreign body, and vocal cord dysfunction can mimic. |
| Symptoms worse at night/early morning               | Airway tone and inflammation vary circadianly | Nocturnal cough alone can be asthma, GERD, postnasal drip, or ACE inhibitor effect.                       |
| FEV1/FVC low and improves after bronchodilator      | Variable obstruction                          | Normal spirometry between attacks does not rule out asthma.                                               |
| Peak flow drops at work and improves away from work | Occupational asthma signal                    | Ask about job tasks, cleaning agents, isocyanates, flour, animals, metals.                                |
| Nasal polyps + asthma + NSAID reaction              | Aspirin-exacerbated respiratory disease       | Avoid nonselective NSAIDs; consider leukotriene pathway therapy and specialist care.                      |

## 10. Chronic asthma management

Modern asthma management emphasizes anti-inflammatory therapy. A key concept is that symptoms can improve rapidly with bronchodilation while airway inflammation remains active; relying only on a short-acting beta agonist can leave the patient at risk for severe exacerbation. Current strategy documents emphasize that adults and adolescents with asthma should receive ICS-containing treatment, either regularly or as-needed in combination regimens, rather than SABA-only treatment. The exact regimen depends on symptom frequency, exacerbation risk, access, adherence, and local guideline adoption.

Inhaled corticosteroids reduce airway inflammation, improve control, and reduce exacerbation risk. LABAs should not be used alone in asthma; they are paired with ICS. Leukotriene receptor antagonists may help allergic rhinitis, aspirin-exacerbated respiratory disease, or exercise symptoms, but they are usually less effective than ICS for preventing exacerbations. Severe asthma requires phenotype-driven evaluation: allergic IgE-mediated disease, eosinophilic disease, type 2 inflammation, chronic oral steroid exposure, comorbid nasal polyps, and biologic eligibility. Before escalating to biologics, confirm the diagnosis, inhaler technique, adherence, and modifiable triggers.

| Medication class | Examples                | Clinical role                                              | Safety/testing association                                         |
|------------------|-------------------------|------------------------------------------------------------|--------------------------------------------------------------------|
| SABA             | Albuterol               | Rapid symptom relief; acute bronchospasm                   | Overuse signals poor control and exacerbation risk.                |
| ICS              | Budesonide, fluticasone | Core anti-inflammatory controller                          | Rinse mouth to reduce dysphonia/thrush; use lowest effective dose. |
| ICS-formoterol   | Budesonide-formoterol   | Can serve as maintenance and reliever in selected regimens | Formoterol has rapid onset; do not generalize to all LABAs.        |

| Medication class     | Examples                                    | Clinical role                                      | Safety/testing association                                |
|----------------------|---------------------------------------------|----------------------------------------------------|-----------------------------------------------------------|
| LABA                 | Formoterol, salmeterol                      | Add-on controller with ICS                         | LABA monotherapy is contraindicated in asthma.            |
| LAMA                 | Tiotropium                                  | Add-on for uncontrolled asthma                     | Also used in COPD; check diagnosis and inhaler technique. |
| Leukotriene modifier | Montelukast                                 | Allergic rhinitis, exercise symptoms, AERD adjunct | Counsel about neuropsychiatric warning.                   |
| Biologics            | Anti-IgE, anti-IL5/5R, anti-IL4R, anti-TSLP | Severe phenotype-directed asthma                   | Requires specialist phenotype assessment.                 |

## 11. Acute asthma exacerbation

Acute asthma ranges from mild bronchospasm to impending respiratory arrest. Severe attacks show tachypnea, tachycardia, accessory muscle use, inability to speak full sentences, low peak flow, hypoxemia, and sometimes pulsus paradoxus. A normal or rising PaCO<sub>2</sub> during a severe attack is dangerous because early asthma exacerbations usually cause hyperventilation and low PaCO<sub>2</sub>. When PaCO<sub>2</sub> normalizes in a patient who still appears distressed, respiratory muscle fatigue may be developing. Silent chest, altered mental status, exhaustion, cyanosis, or bradycardia are late warning signs.

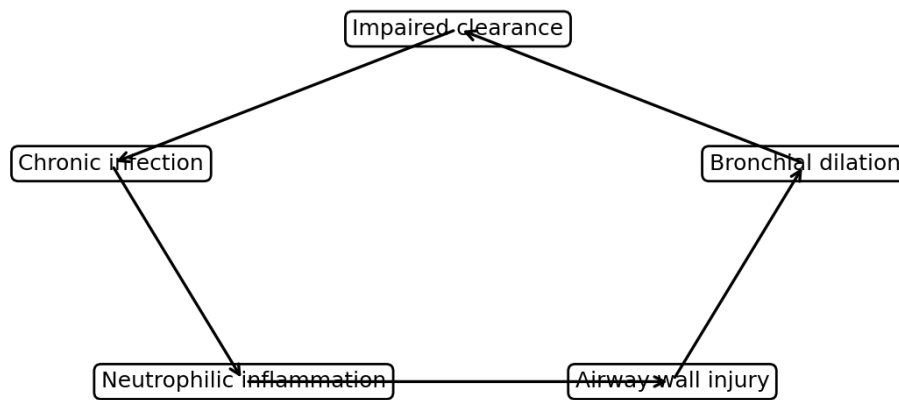
Initial emergency treatment includes inhaled SABA, often combined with inhaled ipratropium for severe exacerbations, systemic corticosteroids, and supplemental oxygen. IV magnesium sulfate is considered in severe exacerbations that do not respond to initial bronchodilators. Antibiotics are not routine unless there is evidence of bacterial infection. Chest x-ray is not required for every asthma attack, but it is useful when pneumothorax, pneumonia, foreign body, heart failure, or atypical features are suspected. Intubation is a clinical decision made before arrest, but it is high-risk because dynamic hyperinflation can worsen hypotension and barotrauma.

**Red flags for ICU-level asthma: prior intubation, inability to speak, altered mental status, silent chest, persistent SpO<sub>2</sub> below goal despite oxygen, peak expiratory flow <25-33% predicted/personal best, rising PaCO<sub>2</sub>, exhaustion, hypotension, or worsening acidosis.**

## 12. Bronchiectasis: illness script and diagnostic approach

Bronchiectasis is permanent abnormal dilation of bronchi caused by chronic airway injury and impaired clearance. The classic script is chronic cough with daily mucopurulent sputum, recurrent respiratory infections, intermittent hemoptysis, crackles or rhonchi, and CT evidence of dilated airways. The disease is both an anatomic finding and a clinical syndrome. Once bronchi become dilated and scarred, secretions pool, bacteria persist, neutrophilic inflammation accelerates injury, and the cycle continues.

## Bronchiectasis cycle: why clearance and cultures matter



Airway clearance + sputum-guided antibiotics attempt to break the cycle.

Figure 2C-2. Bronchiectasis cycle. Chronic infection and inflammation reinforce airway dilation; airway clearance and sputum-guided antibiotics are disease-logic interventions.

High-resolution CT is the diagnostic imaging test. CT may show bronchi that fail to taper, airways wider than adjacent pulmonary arteries, visible peripheral bronchi, bronchial wall thickening, mucus plugging, and tree-in-bud opacities during infection. Workup should search for a cause when clinically appropriate: prior severe infection, aspiration, allergic bronchopulmonary aspergillosis, immunodeficiency, rheumatoid arthritis, inflammatory bowel disease, primary ciliary dyskinesia, alpha-1 antitrypsin deficiency, and cystic fibrosis or CFTR-related disorder.

| Cause category                 | Examples                                                  | Clue that points toward it                                                         |
|--------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------|
| Postinfectious injury          | Severe bacterial pneumonia, TB, pertussis, viral injury   | Localized bronchiectasis after a major infection.                                  |
| Impaired mucociliary clearance | Primary ciliary dyskinesia, Kartagener syndrome, CF       | Chronic sinusitis, otitis, infertility, situs inversus, childhood symptoms.        |
| Immune deficiency              | Common variable immunodeficiency, IgG subclass deficiency | Recurrent sinopulmonary infections, low immunoglobulins.                           |
| Allergic/inflammatory          | ABPA, rheumatoid arthritis, IBD                           | Asthma, eosinophilia, high IgE, central bronchiectasis, systemic autoimmune signs. |
| Obstruction/aspiration         | Tumor, foreign body, chronic aspiration                   | Focal disease, recurrent infection in same lobe, dysphagia/reflux.                 |

## 13. Bronchiectasis management

Bronchiectasis management is built around airway clearance, sputum microbiology, exacerbation prevention, and cause-specific therapy. Airway clearance is not optional education; it is a central treatment because retained secretions drive infection and inflammation. Techniques include active cycle breathing, oscillatory positive expiratory pressure devices, chest physiotherapy, exercise, hydration, and nebulized therapies in selected patients. Bronchodilators are used when there is coexisting asthma, COPD, or demonstrable bronchodilator response; they are not automatically required for all bronchiectasis.

Exacerbations are usually treated with antibiotics guided by prior sputum culture, current severity, and local resistance patterns. A 14-day course is a common standard for significant bronchiectasis exacerbations, especially with *Pseudomonas aeruginosa* or severe disease, although shorter courses may be reasonable in mild, rapidly responding infections depending on organism and clinical context. Chronic macrolide therapy or inhaled antibiotics may be considered in frequent exacerbators after careful evaluation for nontuberculous mycobacteria, QT risk, hearing risk, resistance, and *Pseudomonas* status.

**Hemoptysis is a common complication because inflamed bronchial arteries become fragile. Mild hemoptysis may occur during infection, but massive hemoptysis is an emergency requiring airway protection planning, localization, reversal of coagulopathy, CT angiography or bronchoscopy depending on stability, and possible bronchial artery embolization.**

## 14. Adult cystic fibrosis and CFTR-related disease

Cystic fibrosis is an autosomal recessive disorder caused by pathogenic variants in CFTR, a chloride and bicarbonate channel. Impaired epithelial ion transport dehydrates airway secretions, thickens mucus, reduces mucociliary clearance, and predisposes to chronic infection and bronchiectasis. Although CF is classically diagnosed in childhood, adults increasingly appear in internal medicine because survival has improved and because milder CFTR-related disease can present later with bronchiectasis, chronic sinusitis, pancreatitis, male infertility, or recurrent *Pseudomonas* infection.

Pulmonary disease in CF often progresses from intermittent infection to chronic colonization with organisms such as *Staphylococcus aureus*, *Haemophilus influenzae*, and *Pseudomonas aeruginosa*. Non-tuberculous mycobacteria and *Burkholderia cepacia* complex matter in advanced or specialized care. Extrapulmonary associations include pancreatic insufficiency, fat-soluble vitamin deficiency, diabetes related to CF, liver disease, distal intestinal obstruction syndrome, infertility, and osteoporosis.

| CF domain          | Mechanism                                                  | Clinical association                                     | Management association                                                                                                      |
|--------------------|------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Airways            | Dehydrated mucus and impaired clearance                    | Bronchiectasis, chronic infection, hemoptysis            | Airway clearance, dornase alfa/hypertonic saline in selected patients, inhaled/IV antibiotics, CFTR modulators by genotype. |
| Pancreas           | Duct obstruction and exocrine injury                       | Steatorrhea, weight loss, fat-soluble vitamin deficiency | Pancreatic enzyme replacement, vitamins A/D/E/K, nutrition support.                                                         |
| Endocrine pancreas | Chronic pancreatic damage                                  | Cystic-fibrosis-related diabetes                         | Screening and insulin-centered management when diagnosed.                                                                   |
| Reproductive tract | Congenital bilateral absence of vas deferens in many males | Male infertility                                         | Fertility counseling and assisted reproduction options.                                                                     |
| Sinuses            | Thick secretions and chronic inflammation                  | Chronic sinusitis, nasal polyps                          | ENT care, saline, topical therapy, infection treatment.                                                                     |

## 15. Differentials and mimics of wheeze

Wheezing is airflow through narrowed or compressed airways, not a diagnosis. The same sound can occur in asthma, COPD, bronchiectasis, heart failure with airway edema, anaphylaxis, pulmonary embolism, foreign body aspiration, tracheomalacia, endobronchial tumor, vocal cord dysfunction, and bronchiolitis. The clinician should ask whether the wheeze is episodic or persistent, inspiratory or expiratory, focal or diffuse, associated with urticaria or hypotension, associated with crackles/edema, triggered by exertion or allergens, and responsive to bronchodilator.

| Mimic                     | Clue against simple asthma/COPD                                                    | Next diagnostic move                                                   |
|---------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Heart failure             | Orthopnea, edema, S3, crackles, high BNP, cardiomegaly                             | CXR, BNP, echocardiography, diuretic response.                         |
| Vocal cord dysfunction    | Throat tightness, inspiratory noise, abrupt episodes, poor bronchodilator response | Flow-volume loop, laryngoscopy during symptoms.                        |
| Pulmonary embolism        | Pleuritic pain, tachycardia, risk factors, clear lungs                             | Clinical probability, D-dimer/CTPA as appropriate.                     |
| Anaphylaxis               | Urticaria, hypotension, angioedema, trigger exposure                               | Immediate epinephrine; airway monitoring.                              |
| Endobronchial obstruction | Focal monophonic wheeze, recurrent pneumonia same location                         | CT chest and bronchoscopy.                                             |
| Pneumothorax              | Sudden pleuritic pain, unilateral decreased breath sounds                          | CXR or bedside ultrasound; urgent decompression if tension physiology. |

## 16. High-yield exam clues

| Clue                                                                       | Most likely diagnosis or concept               | Why it matters                                                                      |
|----------------------------------------------------------------------------|------------------------------------------------|-------------------------------------------------------------------------------------|
| Productive cough $\geq 3$ months/year for $\geq 2$ consecutive years       | Chronic bronchitis                             | Clinical definition; may coexist with emphysema.                                    |
| Young nonsmoker with basilar emphysema or liver disease                    | Alpha-1 antitrypsin deficiency                 | Test level/genotype; smoking accelerates disease.                                   |
| COPD with $\text{PaO}_2 \leq 55$ mm Hg or $\text{SaO}_2 \leq 88\%$ at rest | Long-term oxygen therapy indication            | One of the few COPD interventions with survival benefit in severe hypoxemia.        |
| COPD exacerbation oxygen target 88-92%                                     | Avoid hyperoxia-induced hypercapnia            | Too much oxygen can worsen $\text{CO}_2$ retention in susceptible patients.         |
| Asthma + nasal polyps + respiratory reaction to NSAIDs                     | Aspirin-exacerbated respiratory disease        | Avoid triggers; leukotriene pathway and specialty therapy may help.                 |
| Severe asthma with normal or rising $\text{PaCO}_2$                        | Impending respiratory failure                  | Asthma usually lowers $\text{PaCO}_2$ early; normal $\text{CO}_2$ can mean fatigue. |
| Years of foul or purulent sputum with recurrent pneumonia                  | Bronchiectasis                                 | Confirm with high-resolution CT and sputum culture.                                 |
| Bronchiectasis + chronic sinusitis + situs inversus                        | Primary ciliary dyskinesia/Kartagener syndrome | Mucociliary clearance disorder.                                                     |
| Bronchiectasis + pancreatic insufficiency + male infertility               | Cystic fibrosis/CFTR-related disease           | Sweat chloride, CFTR testing, specialized care.                                     |

## 17. Common pitfalls

- Calling every wheeze asthma. Wheeze is a sound; the cause may be heart failure, PE, anaphylaxis, endobronchial tumor, or vocal cord dysfunction.
- Using chronic oral steroids for stable COPD. Systemic steroids are for exacerbations or specific inflammatory indications, not routine long-term COPD maintenance.
- Forgetting that ICS in COPD is selective. Inhaled corticosteroids reduce exacerbations in some COPD patients but can increase pneumonia risk.

- Missing respiratory fatigue in acute asthma because the patient is no longer wheezing loudly. A silent chest, rising PaCO<sub>2</sub>, and altered mentation are late dangerous signs.
- Treating bronchiectasis exacerbations without sputum culture history. Prior organisms, especially *Pseudomonas*, should influence antibiotic choices.
- Ignoring comorbid disease. COPD and asthma outcomes are worsened by GERD, rhinitis, OSA, obesity, anxiety, heart failure, CAD, osteoporosis, and poor inhaler technique.
- Assuming normal pulse oximetry excludes serious disease. PE, early asthma, hypercapnia, and exertional desaturation may require additional assessment.

## 18. Original practice questions

Question 1. A 67-year-old former smoker has progressive exertional dyspnea, decreased breath sounds, hyperinflation on CXR, and post-bronchodilator FEV<sub>1</sub>/FVC of 0.55. Resting SaO<sub>2</sub> is 87% on room air on repeated testing. Which intervention is most likely to improve survival?

Answer. Long-term oxygen therapy, because severe chronic resting hypoxemia in COPD is one of the classic settings where oxygen improves survival. Smoking cessation also improves outcomes, but in this already hypoxemic patient the oxygen threshold is the numeric clue.

Question 2. A 23-year-old has episodic chest tightness and wheeze after running in cold air. Spirometry shows obstruction that improves by 15% and 260 mL after albuterol. What disease feature is being demonstrated?

Answer. Variable reversible airflow limitation, supporting asthma. The numeric threshold used here is at least 12% and at least 200 mL improvement in FEV<sub>1</sub> or FVC.

Question 3. A 58-year-old with COPD is brought in for worsening dyspnea and purulent sputum. ABG shows pH 7.28 and PaCO<sub>2</sub> 68 mm Hg. He is awake but using accessory muscles. What respiratory support should be considered early?

Answer. Noninvasive ventilation, because acute-on-chronic hypercapnic respiratory failure with respiratory acidosis and increased work of breathing is a classic NIV scenario unless contraindications are present.

Question 4. A 42-year-old has chronic daily foul-smelling sputum, recurrent pneumonias, and intermittent hemoptysis. Which imaging test best confirms the suspected chronic airway diagnosis?

Answer. High-resolution CT of the chest, which can show airway dilation, lack of tapering, bronchial wall thickening, and mucus plugging consistent with bronchiectasis.

Question 5. A patient with severe asthma initially has low PaCO<sub>2</sub> from hyperventilation. Two hours later he remains visibly exhausted, but PaCO<sub>2</sub> is now normal. Why is this concerning?

Answer. A normalizing or rising PaCO<sub>2</sub> in severe asthma can indicate respiratory muscle fatigue and impending ventilatory failure, not improvement.

Question 6. A man with bronchiectasis, chronic sinusitis, and infertility is found to have situs inversus. What mechanism connects these findings?

Answer. Primary ciliary dyskinesia causes impaired mucociliary clearance; Kartagener syndrome refers to the association with situs inversus.

## 19. Graph RAG extraction map

The following table converts the chapter into candidate graph nodes and edges. It is not meant as a final ontology; it is a starting structure for concept extraction.

| Node A                                  | Edge type       | Node B                                 | Rationale                                                                                        |
|-----------------------------------------|-----------------|----------------------------------------|--------------------------------------------------------------------------------------------------|
| Tobacco smoke                           | causes/promotes | COPD                                   | Exposure recruits inflammation, oxidative stress, and protease-antiprotease imbalance.           |
| Alpha-1 antitrypsin deficiency          | predisposes to  | Panacinar emphysema                    | Low antiprotease activity permits elastase-mediated alveolar destruction.                        |
| COPD exacerbation                       | treated with    | SABA/SAMA + systemic steroids          | Bronchodilation and anti-inflammatory treatment reduce airflow limitation and treatment failure. |
| Severe chronic hypoxemia                | treated with    | Long-term oxygen therapy               | LTOT improves survival in COPD with severe resting hypoxemia.                                    |
| Asthma                                  | defined by      | Variable airflow limitation            | Symptoms and obstruction fluctuate over time and often improve with bronchodilator.              |
| Aspirin-exacerbated respiratory disease | associated with | Asthma + nasal polyps + NSAID reaction | Triad suggests leukotriene-driven upper/lower airway disease.                                    |
| Bronchiectasis                          | diagnosed by    | High-resolution CT                     | CT demonstrates permanent airway dilation and wall thickening.                                   |
| Impaired airway clearance               | drives          | Bronchiectasis infection cycle         | Retained secretions allow bacterial persistence and neutrophilic injury.                         |
| CFTR dysfunction                        | causes          | Thick airway secretions                | Abnormal chloride/bicarbonate transport dehydrates mucus and impairs clearance.                  |

## 20. References and guideline anchors

Global Initiative for Chronic Obstructive Lung Disease. 2026 GOLD Report and Pocket Guide: Global Strategy for Prevention, Diagnosis and Management of COPD. GOLD; 2026.

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Chalmers JD, et al. European Respiratory Society clinical practice guideline for management of adult bronchiectasis. European Respiratory Journal. 2025.

Cystic Fibrosis Foundation. Adult Care Clinical Care Guidelines; Pulmonary Exacerbations Clinical Care Guidelines; Pancreatic Enzymes Clinical Care Guidelines.

Medvale source-use note: uploaded pulmonology sample PDF and Anki CSV were used only as topic coverage signals. This chapter is an original synthesis with original tables and diagrams.